

Quantum Circuit Learning: QC-ML

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1 QCL Algorithm

The Quantum Circuit - Machine Learning (QC-ML) project will aim to replicate a quantum-classical hybrid system designed to model machine learning. Initially, the project will focus on replicating the QCL algorithm designed by (*K. Mitari, 2018*), which is laid out below:

- Encode input data $\{x_i\}$ into some quantum state $|\psi_{\text{in}}(x_i)\rangle$ by applying a unitary input gate $U(x_i)$ to initialized qubits $|0\rangle$.
- Apply a θ -parameterized unitary $U(\theta)$ to the inout state and generate an output state $|\psi_{\text{out}}(x_i, \theta)\rangle = U(\theta) |\psi_{\text{in}}(x_i)\rangle$.
- Measure the expectation values of some chosen observables. *K. Mitari, 2018* used a subset of Pauli operators $\{B_j\} \subset \{I, X, Y, Z\}^{\otimes N}$. Using some output function F , output $y_i = y(x_i, \theta)$ is defined to be $y(x_i, \theta) \equiv F(\{\langle B_j(x_i, \theta) \rangle\})$.
- Minimize the cost function $L(f(x_i), y(x_i, \theta))$ of the teacher $f(x_i)$ and the output y_i , by tuning the circuit parameters θ iteratively.
- Evaluate the performance by checking the cost function with respect to the data set that is taken independently from the training one.